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Seafloor imagery with an advanced imaging sonar system

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The mapping swath and along-track resolution can be simultaneously enhanced by using multireceiver synthetic aperture sonar (SAS). Unfortunately, the SAS parameters, including the pulse repetition frequency, SAS moving velocity, and the along-track aperture of the receiver array, are rigorously limited. Any deviation from the strict requirements could result in sampling that is not uniform. Based on the matrix inversion (MI) method, the uniform signal is recovered. The MI technique would not be able to recreate the SAS image if the redundant receivers within two subsequent pulses were extremely close or overlapped. This study discusses an innovative approach to tackle this issue. Our approach first identifies the receivers that the SAS image reconstruction algorithms exploit. Then, the approximation error caused by the range approximation is compensated for these receiver datasets. Thereafter, all receiver datasets corresponding to each pulse are stored pulse by pulse. After this operation, the omega-k algorithm is further used to reconstruct the SAS image. When the MI method works well, the proposed method can achieve high-quality images that closely match those of the matrix inversion method. More importantly, the proposed method can still work in cases where the MI method fails. The experiments further show the major superiority of our method over traditional methods.

KEYWORDS

imaging sonar, non-uniform sampling, omega-k algorithm, seafloor imagery, synthetic aperture sonar

1 Introduction

The target detection (Huang et al., 2023; Zhou et al., 2025; Han et al., 2025; Li et al., 2024) and seafloor mapping (Huang and Yang, 2022; Zhang et al., 2022a; Mao et al., 2025) are important aspects of ocean observation and underwater engineering fields (Su et al., 2025; Xu et al., 2025a, 2026, 2025b; Mwaffo, 2024). Owing to its outstanding resolution, synthetic aperture sonar (SAS) (Zhang et al., 2022b, 2026a, 2026b), which is based on the synthetic aperture technique, is a useful instrument for undersea engineering (Ghaban et al., 2024; Zhang et al., 2019; Huang et al., 2022) and ocean observation (Huang et al., 2021, 2019; Zhang et al., 2017b; He et al., 2025). Currently, underwater tasks such as underwater imagery and mapping (Shouldice and Myers, 2023; Zhang and Yang, 2019; Zhang et al., 2024; Guan et al., 2026), underwater target detection (Barbary and Elazeem, 2024; Du et al.,

2024; Zhang et al., 2025), underwater search and rescue (LeHardy and Larsen, 2015; Zhu et al., 2024; Zhang et al., 2018a), underwater localization (Huang et al., 2026; Liu et al., 2025; Ye et al., 2025; Qiu and Guo, 2025; Kim, 2024), underwater tracking (Chen et al., 2025; Gong et al., 2025; Bi et al., 2025; Li et al., 2025; Liang et al., 2025), countermeasures for unexploded ordnance (Williams, 2011; Hall et al., 2019; Malkasse et al., 2023), and underwater archaeology (Ferentinos et al., 2020; Tan et al., 2019; Marston et al., 2024) all greatly benefit from the high-resolution SAS system. To date, the SAS system exploits multiple receivers along the moving direction to enhance the mapping swath and along-track resolution simultaneously, and this configuration is referred to as a multireceiver SAS (Marston et al., 2023; Zhang et al., 2023). When this system is employed, all elements are configured as a linear receiving array. The synchronization corresponding to all elements is achieved via sampling circuits, which are further controlled by a field programmable gate array (FPGA) using the same sample wiring. However, the relationships among the SAS parameters, such as the pulse interval frequency (PRF), SAS moving velocity, and the along-track aperture of the receiver array, are rigorously limited (Krieger et al., 2004; Zhang and Yang, 2021; Wang and Huang, 2023). In summary, the forward distance within each pulse should be exactly half the length of the receiver array. The uniform signal can only be achieved in this case, where traditional SAS imaging algorithms can be directly used. In practice, ocean conditions (Zhang et al., 2018b; Guo et al., 2025; Fei et al., 2025) are significantly affected by factors such as wind, wave, and ocean currents. This means that the SAS system cannot move the expected distance during each transmitted pulse. In other words, the non-uniform sampling (Yang and Liu, 2022) in the along-track dimension arises from the inability to consistently maintain rigorous restrictions among the SAS parameters (Zhang et al., 2026d, 2026c).

In the work of Wang et al. (2019), the omega-k algorithm is discussed based on an exact analytical two-dimensional (2D) spectrum. It requires Stolt mapping for each receiver dataset. The Stolt mapping highly depends on the interpolation in the 2D frequency domain. In practice, the coherent superposition in the 2D frequency domain after interpolation cannot guarantee imaging accuracy. Moreover, the method in that study can only work when SAS parameters satisfy the strict relationships. In order to relax the strict relationships, the uniform signal is reconstructed using non-uniform datasets, and then the omega-k algorithm is applied. In the study by Zhang et al. (2009), the spectrum of a non-uniform signal can be formulated as the spectrum weighting of a uniform signal. The weighting coefficients are independent of the frequency. Therefore, the spectrum of a uniform signal can be recovered after calculating the weighting coefficients. Nevertheless, it is required to compute the weighting coefficients for every along-track frequency. As a result, this approach is characterized by a considerable computational burden. The sampling in the along-track dimension is carried out based on the array of receiver sensors. In view of this factor, the along-track sampling has a periodic structure. Based on this fact, the weighting coefficients are decomposed into a coefficient matrix, which is independent of the input signal (Xiang et al., 2023; Wang et al., 2015). Accordingly, the coefficient matrix can be calculated first, and the reconstruction of a

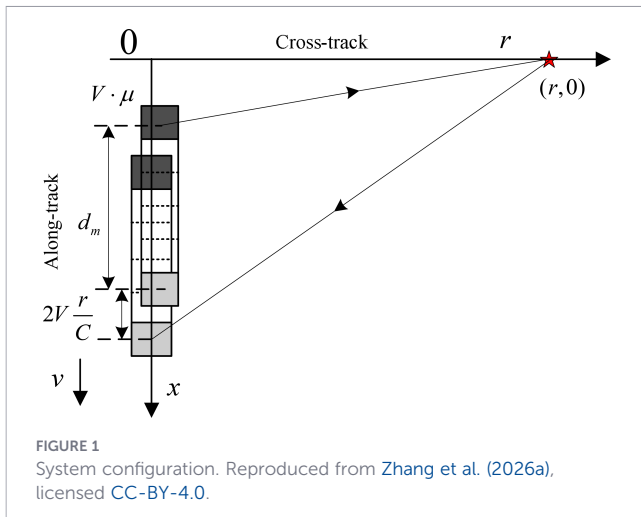
uniform signal would be considerably faster. Based on the theory of multichannel sampling (Brown, 1981; Papoulis, 1977), the multireceiver SAS signal is approximately reformulated as the time delay of the reference receiver echo. According to relationships between the multireceiver SAS signal and the reference receiver, the transfer function matrix can be constructed. The reconstruction matrix is generated through the matrix inversion computation of the transfer function (Yang et al., 2007). Due to this reason, this approach is called the matrix inversion (MI) method. Using each receiver spectrum and the reconstruction matrix, the uniform signal can be easily reconstructed. Although these methods can reconstruct the uniform signal using the non-uniform datasets, they still suffer from two major drawbacks. First, these methods should calculate the inversion of the matrix, which needs a large computation load. Second, the calculation of the matrix inversion requires a full-rank matrix. The multireceiver SAS system may work at the undesired velocity, and the overlapping receivers between adjacent pulses would be introduced. Usually, these overlapping receivers are called redundant receivers. When there are no redundant receivers between adjacent pulses, the transfer function matrix satisfies the requirement of full rank. Consequently, it is easy to perform the inversion calculation of the transfer function matrix. When redundant receivers are introduced, the transfer function matrix with full rank cannot be obtained. Thus the inversion calculation in terms of the transfer function matrix cannot be carried out, and it would fail to recover the uniform signal. In general, both drawbacks would be harmful for the uniform signal reconstruction of a multireceiver SAS system, affecting both efficiency and performance.

To address these issues, this research proposes an alternative approach that is both efficient and effective. With our method, the receivers used for the multireceiver SAS imagery are first selected. The receiver datasets are aligned in order pulse by pulse. Then, the receiver and transmitter are approximated by a supposed element in the middle location. The error caused by this assumption is compensated through multiplication and interpolation. After this operation, the omega-k algorithm is adopted to reconstruct the SAS image. Compared with traditional methods, the provided solution can better focus all targets. More importantly, our method can further work in the case where traditional methods fail to reconstruct the SAS image. Thus, our method has broader applicability than traditional approaches.

The framework of our investigation is listed below. Section 2 clarifies the system model. The recommended approach is discussed in Section 3. Some focusing results are exhibited in Section 4. The conclusion of the work is offered in the final section.

2 Descriptions of the system model

First and foremost, the system model is outlined in this section. Figure 1 illustrates the SAS geometry, where the r -axis and x -axis denote the cross-track and along-track directions, respectively. The traveling speed of the sonar system is denoted by the variable V . The speed related to the underwater acoustic is represented by the constant variable C .



The transmitter, which is shown as a black rectangle, emits the chirp waveform $s(\eta)$. Then, the receivers, denoted by other rectangles, receive the echo (Zhang, 2024), and the echo signal is expressed as Equation 1.

$$s_m(\eta, \mu) = s\left(\eta - \frac{L_m(\mu; r)}{C}\right) \cdot \exp\left\{-j2\pi v_c \frac{L_m(\mu; r)}{C}\right\} \quad (1)$$

The receiver that is closest to the transmitter is marked as the first one. The final receiver is identified as the M th one. The array consists of M receivers. The center frequency is indicated by v_c . The slow time and fast time, corresponding to the along-track and cross-track directions, are denoted by the variables μ and η , respectively. The closest range of the target is indicated by r . The total distance between the sonar and the object is given by:

$$L_m(\mu; r) = \sqrt{r^2 + V^2\mu^2} + \sqrt{r^2 + (V\mu + d_m + 2V\frac{r}{C})^2} \quad (2)$$

Here, d_m indicates the along-track baseline from the transmitter to the m th receiver. The expression $2r\frac{V}{C}$ illustrates the traveling range when the stop-and-hop assumption (Yang, 2024; Zhang et al., 2017a) is not adopted in this paper.

3 Proposed method

3.1 Focusing algorithm

The traditional MI method (Yang et al., 2007) cannot work in cases where there are overlapped receivers between two adjacent pulses. To address this issue, this section presents a novel method. With our method, the receivers used by the imaging algorithms are chosen based on the PRF, sonar velocity, and receiver aperture in the along-track direction. This relationship is given by Equation 3:

$$N = \text{round}\left\{\frac{2V}{D \times \text{PRF}}\right\} \quad (3)$$

In Equation 3, the $\text{round}\{\cdot\}$ denotes the calculation result, which is rounded to the nearest integer. Here, the variable

$N = M - K$ denotes the number of receivers that are not overlapped between two adjacent pulses.

The series approximation approach in terms of Equation 2 is employed, and the result can be expressed as:

$$L_m(\mu; r) \approx 2\sqrt{r^2 + (V\mu + V\frac{r}{C} + 0.5d_m)^2} + \frac{(V\frac{2r}{C} + d_m)^2}{4r} \quad (4)$$

Based on Equation 4 and the phase-stationary theory, the 2D spectrum of the SAS echo is written as:

$$S_n(v_\eta, v_\mu; r) = S(v_\eta) \cdot \exp\left\{-j\pi v_c \frac{(2\frac{V}{C}r + d_n)^2}{2Cr}\right\} \cdot \exp\left\{-j\pi v_\eta \frac{(2\frac{V}{C}r + d_n)^2}{2Cr}\right\} \times \\ \exp\left\{j\pi v_\mu \frac{d_n}{V}\right\} \cdot \exp\left\{-j4\pi \frac{r}{C} \sqrt{(v_c + v_\eta)^2 - v_\mu^2 \frac{C^2}{4V^2}}\right\} \cdot \exp\left\{j2\pi v_\mu \frac{r}{C}\right\} \quad (5)$$

Employing the numerical spectrum, the errors of the presented spectrum shown in Equation 5 are analyzed. The sonar works with a 0.15 MHz central frequency and 0.02 MHz bandwidth. The along-track aperture of the transmitter is 8 cm. The along-track aperture of the receiver is 4 cm, and the array consists of 32 receivers. The PRF of the emitted chirp is 5 Hz. The motion speed of our sonar is 3.2 m/s. When the distance of the first receiver–transmitter pair is 4 cm, the errors are shown in Figure 2. When the range is 15 m, the error in Figure 2a is approximately 0.005 rad. The error becomes larger with the object range. In Figure 2c, the large error is approximately 0.05 rad for the far object. When the distance of the last receiver–transmitter pair is 128 cm, the errors are shown in Figure 3. The error in Figure 3a is approximately -0.15 rad for the close object. The error decreases as the object range increases. In Figure 3c, the maximum error is approximately 0.05 rad. Based on Figure 2a and Figure 3a, the error of a close object strongly depends on the distance of the transmitter–receiver pair. However, the error of the far object weakly depends on the distance of the transmitter–receiver pair. Overall, the error of Equation 5 is small, and it can be used for the seafloor imagery.

In the frequency domain of the cross-track direction, the compression in the cross-track direction is executed for each receiver echo. The processing filter corresponding to this step is supplied by:

$$H(v_\eta) = \text{conj}\{S(v_\eta)\} \quad (6)$$

Here, the conjunction is represented by $\text{conj}\{\cdot\}$ in Equation 6.

To completely eliminate the impact of the second item in Equation 5, the phase correction is executed in the 2D space domain, and the correction term is given by Equation 7:

$$H_n(\eta, \mu; r) = \exp\left\{j\pi v_c \frac{(2\frac{V}{C}r + d_n)^2}{2Cr}\right\} \quad (7)$$

Subsequently, we remove the influence of the third term in Equation 5 via interpolators, and the corresponding correction term of time delay is shown as Equation 8:

$$\Delta\tau_n(\eta, \mu; r) = \frac{(2\frac{V}{C}r + d_n)^2}{4Cr} \quad (8)$$

In general, the sinc function is usually employed to reconstruct the signal. The reconstructed signal $f(r)$ based on the sampled data $Z(n)$ is shown as:

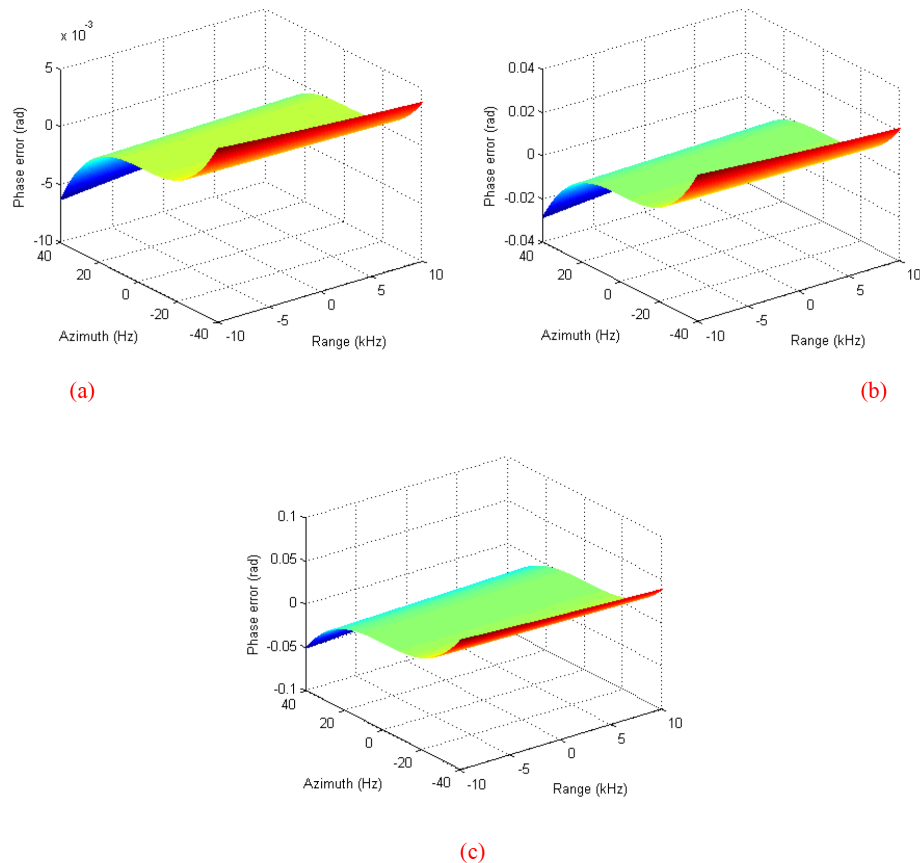


FIGURE 2

Error when the distance of the first receiver–transmitter pair is 4 cm. (a) object range is 15 m, (b) object range is 75 m, and (c) object range is 135 m.

$$f(r) = \sum_{n=-4}^4 Z(n) \frac{\sin[\pi(r-n)]}{\pi(r-n)} \quad (9)$$

In Equation 9, the kernel function is represented by the sinc function, i.e., $\frac{\sin(\pi y)}{\pi y}$. In addition, the kernel length is 8.

All receiver datasets are aligned in order pulse by pulse. The datasets corresponding to N receivers within the first pulse are first stored. In practice, the interleaved storage of all datasets is carried out. As an example, assume that three receiving elements are included in the array. When the sampling of the first pulse is completed, we store the data from these three receivers in order. The subsequent storage of the data from the three receivers after the sampling of the second pulse is then carried out. The storage of sampling data for each pulse is done based on the same procedure. Thereafter, the datasets corresponding to N receivers within the second pulse are stored in order. N receiver datasets within other pulses are stored in order. The flowchart of this procedure is shown in Figure 4. After this operation, the fourth term in Equation 5 is corrected.

The omega-k algorithm is characterized by high performance. In general, this method contains two important operations. The first operation is the reference multiplication based on the reference objects. This performs the buck imaging for the reference objects. The second operation is the Stolt mapping. This performs the differential imaging for the other objects.

Following the alignment of the SAS data, coordinate transformation with respect to the instantaneous frequency is conducted using the interpolation technique known as Stolt mapping. The transformed coordinate is shown as Equation 10:

$$v_r = \sqrt{(v_c + v_\eta)^2 - v_\mu^2 \frac{C^2}{4V^2}} - v_c \quad (10)$$

In the Doppler domain of the along-track direction, the impacts of the final term in Equation 5 are removed. The filter used for phase correction is written as:

$$H_a(\eta, v_\mu; r) = \exp\left\{-j2\pi v_\mu \frac{r}{C}\right\} \quad (11)$$

In the along-track direction, the inverse Fourier transform (IFT) is executed to yield the SAS image. Figure 5 presents a schematic overview of the described strategy explained in this section.

3.2 Discussion of the calculation amount

The computation amount is evaluated in this section. Taking into account the echo from each receiver, the cross-track FT, multiplication, and interpolation are performed twice, twice, and once, respectively. After data alignment, the cross-track FT is performed twice, and the along-track FT is also performed twice.

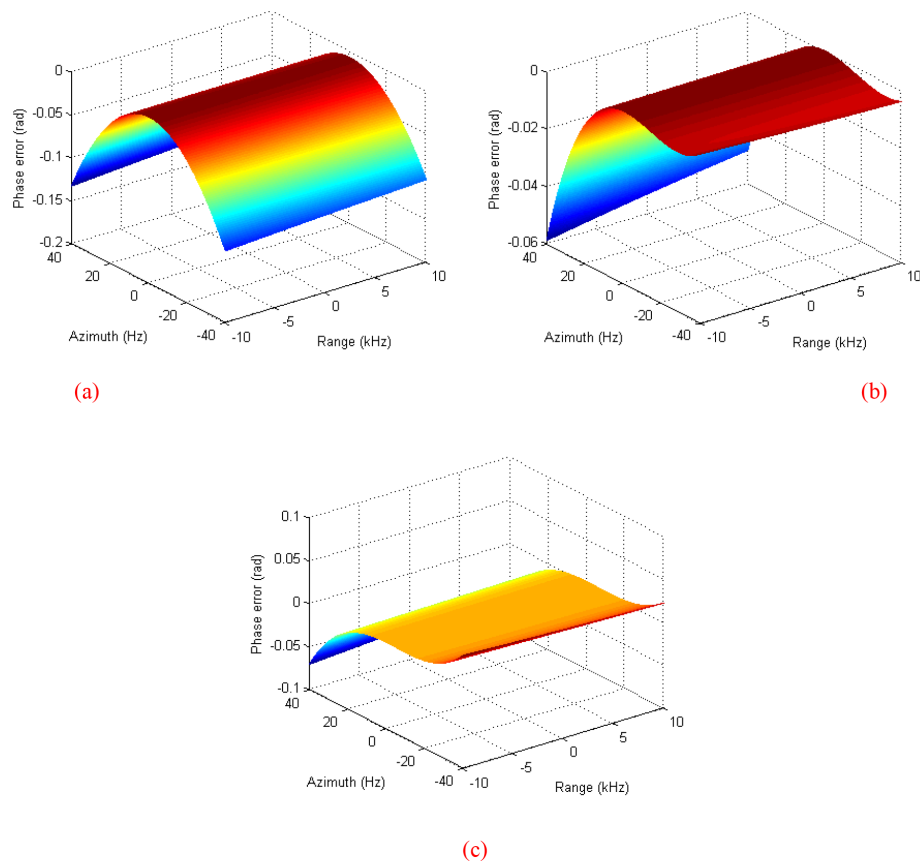


FIGURE 3

Error when the distance of the first receiver–transmitter pair is 128 cm. (a) object range is 15 m, (b) object range is 75 m, and (c) object range is 135 m.

In addition, the interpolation called the Stolt mapping is further applied. For clarity, N_a expresses the number of samples in the along-track direction, while N_r represents the number of samples in the cross-track direction. Furthermore, the length of the sinc interpolator kernel is denoted by M_{ker} . In the work by Cumming and Wong (2005), the computation amount for FT, multiplication, and interpolation were analyzed in detail. The computation amount for the FT/IFT with N_r samples is $5N_r \log_2 N_r$. The computation amount for the multiplication is 6. Getting a single interpolated value requires a computation amount of $2(2M_{\text{ker}} - 1)$. This analysis allows us to figure out the computation amount of our approach,

and it is expressed as:

$$O = 20N_a N_r \log_2 N_r + 12N_a N_r + 4(2M_{\text{ker}} - 1)N_a N_r + 10N_a N_r \log_2 N_a \quad (12)$$

There are 5,000 samples and 7,200 samples in the azimuth and range dimensions, respectively. The interpolator kernel length is 8. Based on Equation 12, the computation amount is approximately 16.3 giga floating-point operations.

4 Experimental results and analysis

4.1 Imaging results based on simulations

We now present the numerical simulation experiments. In our simulation, we mainly discuss the focusing of two ideal targets. The cross-track positions of both objects are 35 m and 115 m. Because both objects are situated at different places, they are labeled as the close and far targets, respectively. In the along-track direction, the corresponding positions of both objects are 3.5 m and 8 m, respectively. The sonar operates at a central frequency of 0.15 MHz with a bandwidth of 0.02. The along-track aperture of the transmitter is 8 cm. The along-track aperture of the receiver is 4 cm, and the array consists of 32 receivers. The PRF of the emitted chirp is 5 Hz. The motion speed of our sonar is 3.15 m/s.

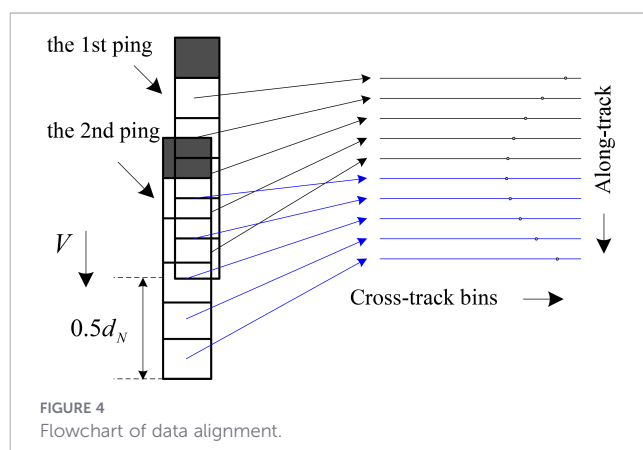


FIGURE 4

Flowchart of data alignment.

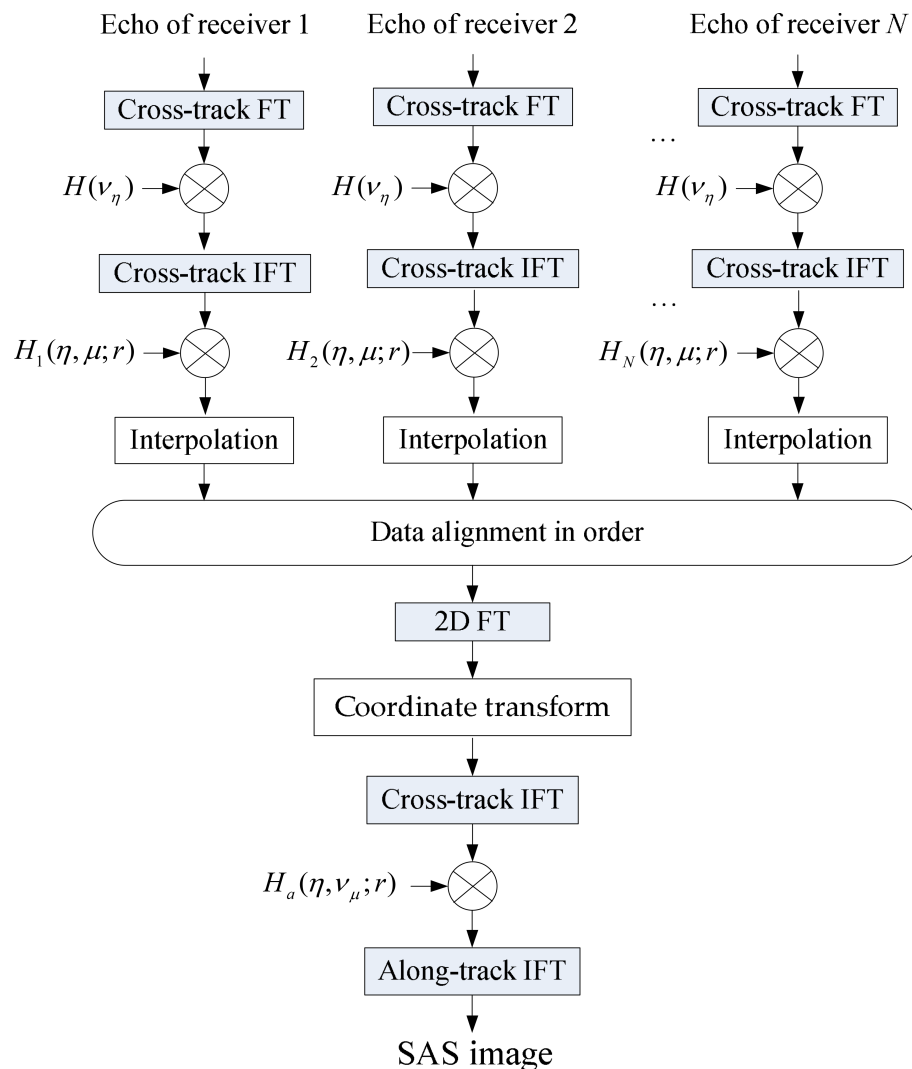


FIGURE 5
Schematic .

Our primary simulation discusses the case where the traditional MI method (Yang et al., 2007) works with the high-quality results. When the system operates at 3.2m/s, all receivers are used to recover the SAS image. When the system operates at 3.1m/s, the imaging algorithm requires only 31 receivers, according to Equation 11. Therefore, a velocity of 3.15m/s is selected, under which the traditional MI method (Yang et al., 2007) can operate effectively in the first simulation case.

Based on the MI method (Yang et al., 2007), Figure 6a shows the filtered output of the MI technique. With the discussed method, the reconstructed images are presented in Figure 6b. The images relying on the back-projection (BP) algorithm (Zhang and Yang, 2022), which serve as the benchmark, are shown in Figure 6c. A comparison of the filtered results of the three methods demonstrates that the proposed approach yields images with remarkable performance, which are comparable to those of the MI technique and BP algorithm. We conclude that the images relying on the proposed method can be comprehensively used without any noticeable loss of imaging performance.

In the along-track direction, Figure 7 presents representative image slices that allow a more detailed comparison of the different approaches. The curves of our method are mostly identical to the

slices generated by the MI method (Yang et al., 2007) and the BP algorithm (Zhang and Yang, 2022). This observation implies that the proposed strategy can achieve remarkable outcomes in cases when the traditional MI method (Yang et al., 2007) operates successfully. This is valuable for practical SAS imaging applications.

The performance can be measured by the indicators. The resolution in the along-track direction (RAT) is defined by the main lobe of the azimuth slice. The peak sidelobe level ratio (PSLR) and integrated sidelobe level ratio (ISLR) are defined in Equation 13 and Equation 14, respectively.

$$\text{PSLR} = 10 \times \lg \frac{Q_{\text{peak}}}{Q_{\text{main}}} \quad (13)$$

$$\text{ISLR} = 10 \times \lg \frac{A - B}{B} \quad (14)$$

In Equation 13, Q_{peak} is the peak intensity of the side lobe. Q_{main} is the peak intensity of the main lobe. In Equation 14, A shows the integrated intensity corresponding to the azimuth slice. B is the integrated intensity of the main lobe.

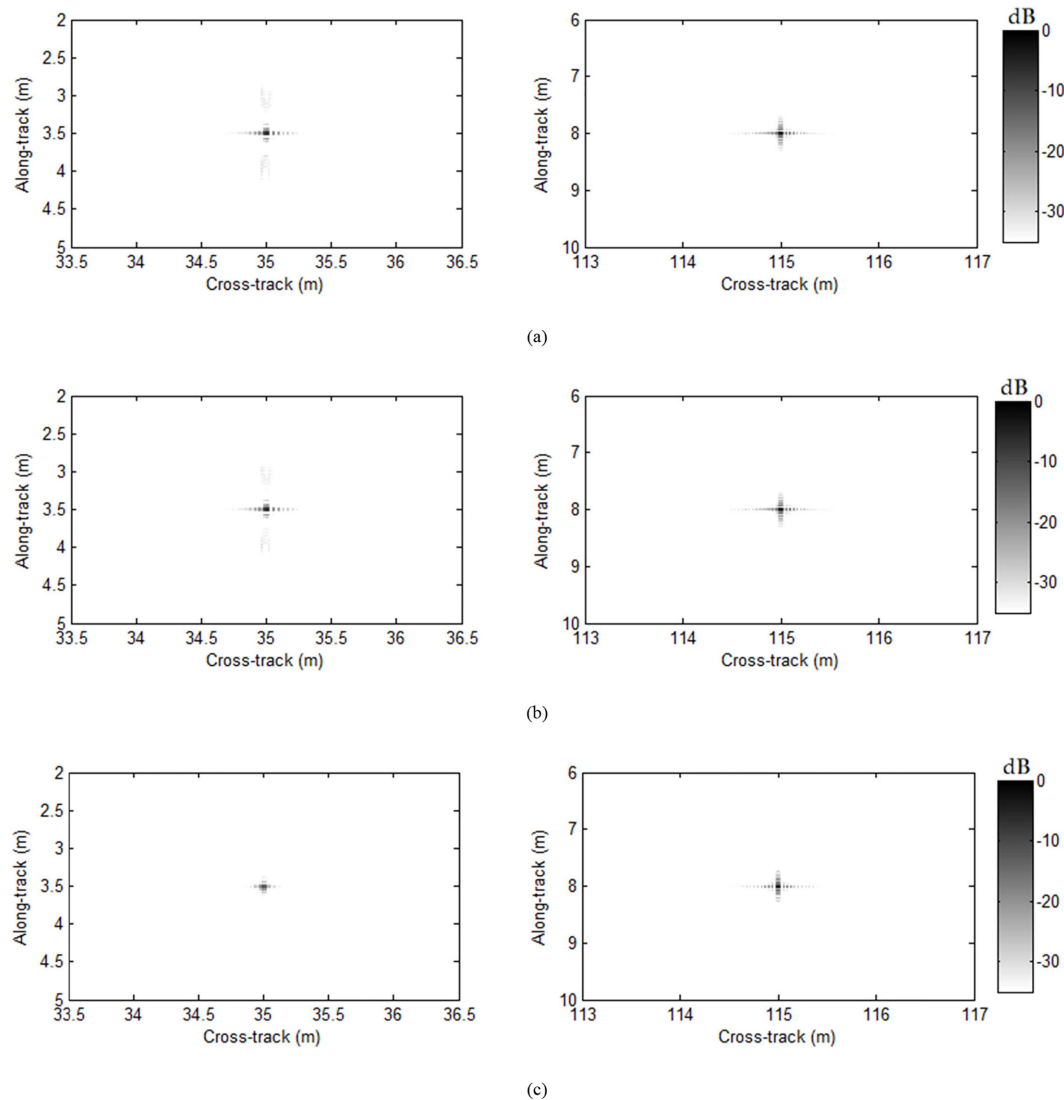


FIGURE 6
Imaging results for the first simulation scenario. (a) MI method, (b) proposed method, and (c) the BP algorithm.

Table 1 summarizes the imaging metrics, including the resolution in the along-track direction (RAT), peak sidelobe level ratio (PSLR), and integrated sidelobe level ratio (ISLR). The calculated metrics indicate that there is little difference between our solution and the MI technique. This demonstrates that our method can also perform well in cases where the traditional MI method (Yang et al., 2007) performs successfully. Therefore, the conclusions drawn from Table 1 are consistent with those obtained from Figures 6 and 7.

The focusing times of the traditional method, proposed method, and the BP algorithm are 382 s, 76 s, and 3560 s, respectively. These results show that the proposed method is very efficient.

We next consider the other simulations. In this simulation, the sonar velocity is 2m/s, while all other SAS parameters remain the same as those used in the previous simulation. With this simulation, there are 20 receivers used for the SAS imagery, according to Equation 11. In other words, there are 12 redundant receivers. Based on the MI method (Yang et al., 2007), the images cannot be recovered as the system matrix of transfer functions with full rank is not acquired. The focusing images shown in Figure 8a confirm this

conclusion. Based on our method, the images with high quality can be recovered successfully. Figure 8b presents the filtering outcome using the proposed solution. These comparisons confirm that the proposed technique can yield high-quality imaging results.

To further evaluate the imaging performance of various techniques, Figure 9 displays the slices in the along-track direction to analyze. Figure 9 reveals that the MI technique (Yang et al., 2007) is unable to generate high-quality well-focused images, and its processing results are significantly deteriorated. In contrast, the targets are reconstructed successfully using our proposed method. This further strengthens the conclusions drawn from Figure 8.

The imagery indicators, i.e., PSLR, ISLR, and RAT, are shown in Table 2. From Table 2, it is clear that the MI method (Yang et al., 2007) cannot reconstruct the SAS image. With our method, the SAS image can be reconstructed successfully. These results indicate that our method can reconstruct the SAS image even when the traditional MI method (Yang et al., 2007) fails.

Taking into account the MI method, the proposed approach, and the BP algorithm, the imaging times are 401 s, 81 s, and 3712 s,

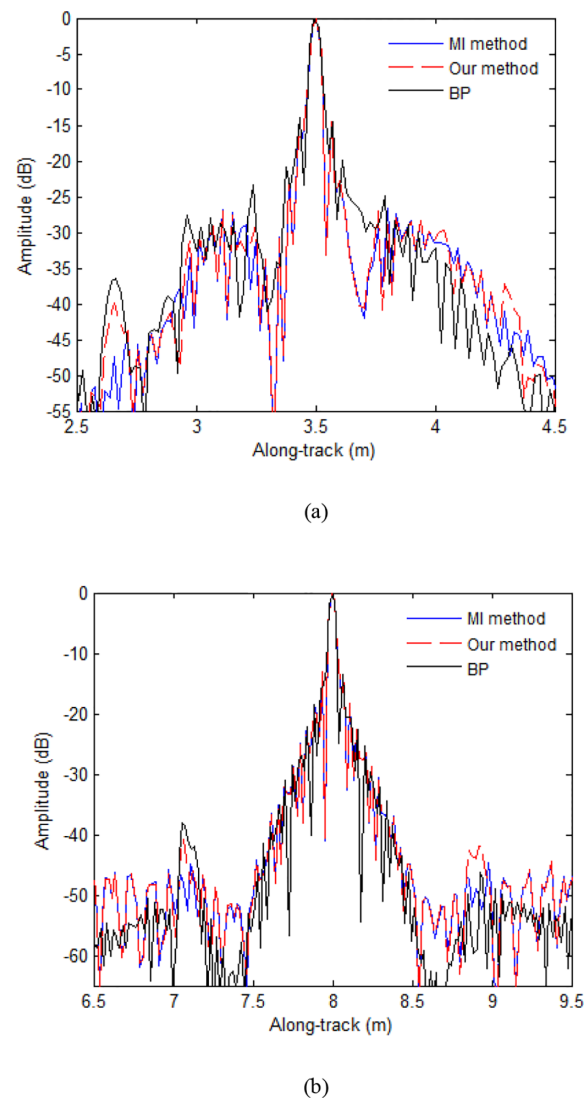


FIGURE 7
Along-track slices for the first simulation scenario. (a) close target and (b) far target.

respectively. The processing time could be significantly saved via the proposed approach.

Based on the simulations in this section, the traditional MI method (Yang et al., 2007) fails to work when redundant receivers are introduced. In contrast, our method performs effectively regardless of whether redundant receivers are introduced.

4.2 Real-data imaging

This section demonstrates the advantages of the proposed method compared to the traditional method using lake-trial data. The performance of the discussed approaches is evaluated using datasets collected by a real SAS system. The sonar velocity is 2.5 m/s. The receiver array possesses 48 receivers. The PRF is 3.125 Hz. The receiver aperture in the along-track direction is 4 cm. According to Equation 11, only 40 receivers are used by the proposed algorithm, leaving 8 redundant receivers that are not used by our method. Figures 10a–c represent the recreated images utilizing the proposed technique, the typical MI method (Yang et al., 2007), and the BP algorithm, respectively.

Comparison of the recovered images associated with the three methods (as seen in Figure 8) demonstrates that the traditional MI method fails to reconstruct the SAS image as the full-rank transfer-function matrix cannot be obtained. In contrast, the proposed method can successfully reconstruct the SAS image, producing results that closely resemble those obtained by the BP algorithm. Compared with the image generated using the MI method, the image quality is significantly improved by applying our method. The experiment using lake-trail datasets further demonstrates the superiority of our method over the traditional MI method (Yang et al., 2007).

To evaluate the efficiency of the three techniques using real SAS datasets, Table 3 summarizes the total computation times of the three imaging techniques. By operating the focusing for each pixel, the BP algorithm can generate high-quality images. Although this method is time-consuming, the results are often regarded as a benchmark. Table 3 shows that the proposed technique is much more effective than the typical MI method (Yang et al., 2007). The conventional MI approach (Yang et al., 2007) demands the computation of the system matrix inversion for each Doppler frequency. Due to this reason, it suffers

TABLE 1 Imagery metrics for the first simulation scenario.

Methods	Close target			Far target		
	PSLR(dB)	ISLR(dB)	RAT (m)	PSLR (dB)	ISLR (dB)	RAT (m)
MI method	-14.59	-9.95	0.04	-13.42	-10.08	0.04
Our method	-14.48	-9.91	0.04	-13.43	-10.06	0.04
BP algorithm	-13.75	-9.98	0.04	-13.43	-10.12	0.04

from real computation load. The proposed method completely avoids matrix inversion and requires only multiplication and interpolation. These factors contribute to the improved efficiency of the proposed method.

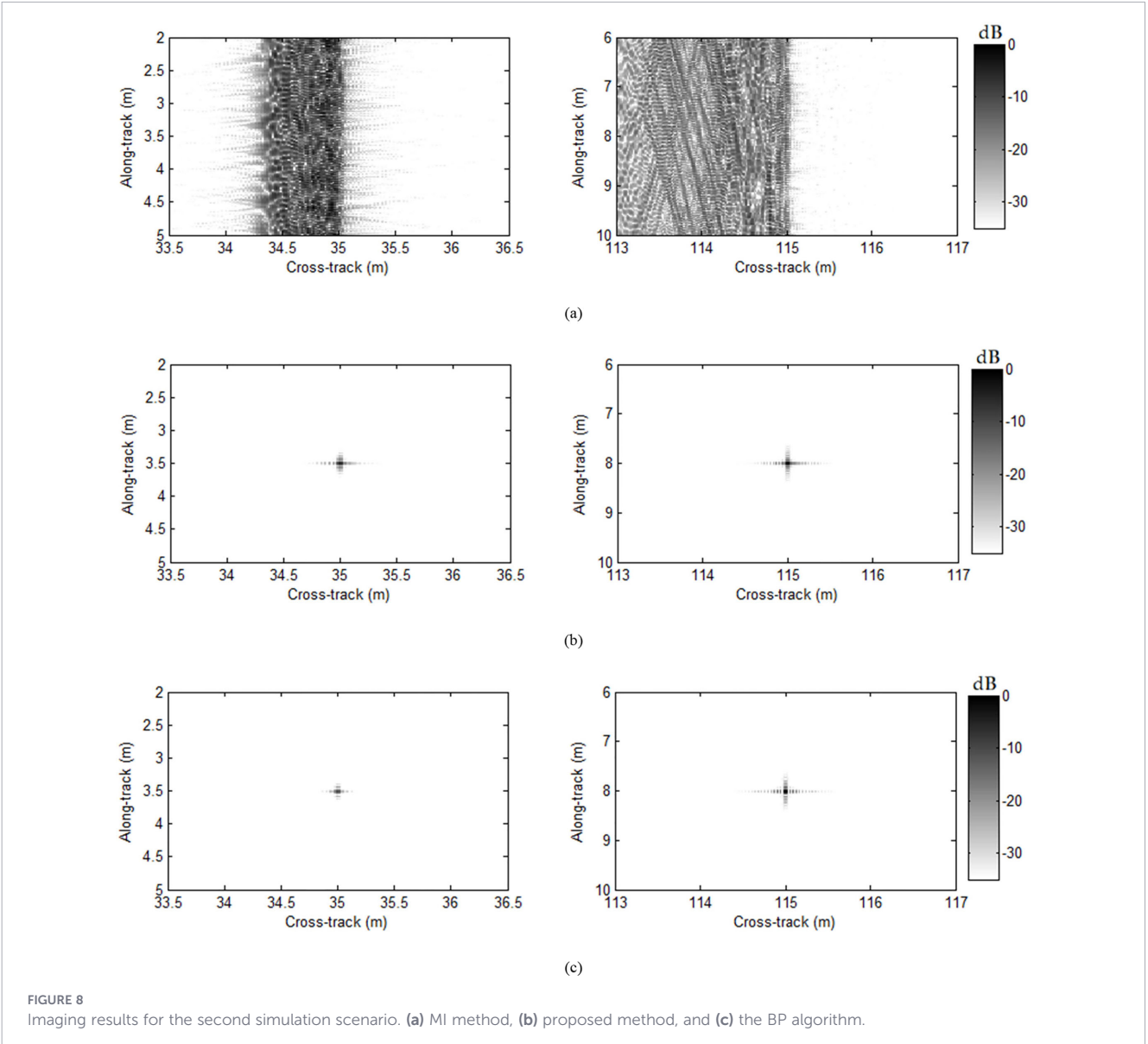
Table 4 presents the contrast (Luo et al., 2024) and entropy (Luo et al., 2024) calculated from real SAS equipment. Both indicators are defined in Equation 15 and Equation 16 respectively.

$$EN = \sum_{k=1}^K \bar{I}(y_k) \ln(\bar{I}(y_k))$$

(15)

$$CO = \frac{\sqrt{\frac{1}{K} \sum_{k=1}^K [I(y_k) - \frac{1}{K} \sum_{k=1}^K I(y_k)]^2}}{\frac{1}{K} \sum_{k=1}^K I(y_k)}$$

(16)



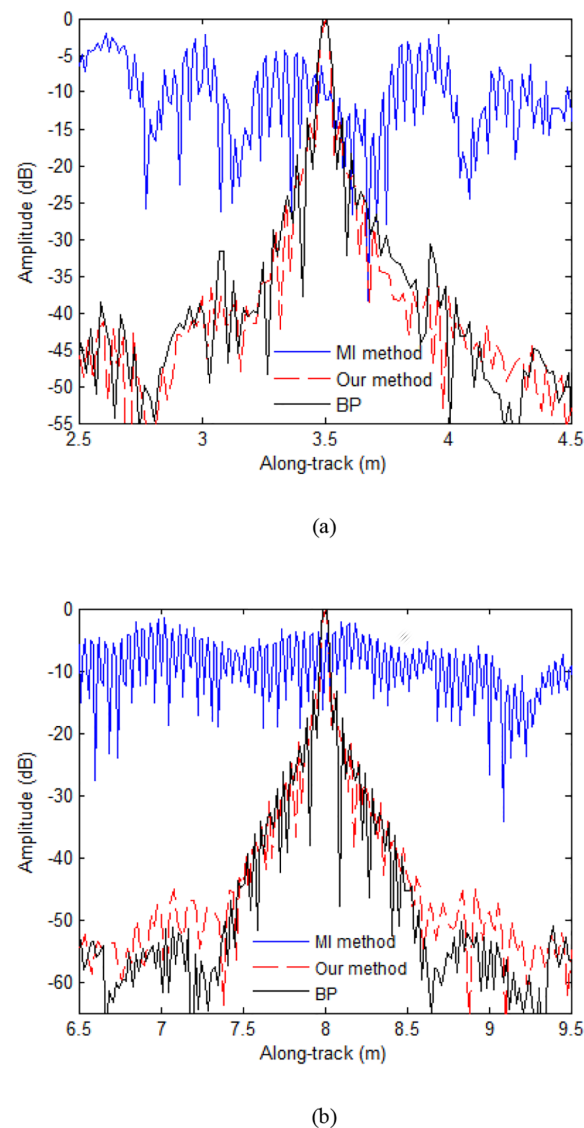


FIGURE 9
Along-track slices for the second simulation scenario. (a) close target and (b) far target.

In Equation 15, the entire sample of images is denoted by K . The k -th sample is represented by $I(y_k)$. The normalized sample is indicated by $\bar{I}(y_k) = \frac{I(y_k)}{\sum_{l=1}^L I(y_l)}$.

As shown in Table 4, the contrast obtained using the MI method is significantly lower than that achieved by the proposed approach. The entropy of the MI method is higher than that of our method. Notably, both parameters using our method are similar to the parameters obtained with the BP algorithm. These results indicate that our method can generate high-quality results compared with the traditional MI method.

5 Conclusions

In this paper, we discuss the multireceiver SAS omega-k algorithm, which has three improved aspects compared with the

traditional MI method. First, the proposed method can generate high-quality imaging results in scenarios where the traditional MI method can operate successfully. Second, the proposed method successfully reconstructs the SAS image even when the conventional MI method cannot obtain the full rank system matrix and is unable to focus the targets. Third, the proposed method avoids computing the inversion of the system matrix, which is required for all Doppler frequencies in the MI technique. As a result, the problems associated with the traditional MI method, such as performance distortions, false objects, and heavy computation load, have been successfully addressed by the proposed method. The proposed method outperforms the traditional MI method. The proposed method can obtain similar performance as the BP algorithm. Since the proposed method avoids the matrix inversion manipulation, the proposed method is very efficient. The simulations results and experiments using lake-trail datasets support these conclusions.

With our current work, the SAS is often working in depth, which is nearly close to the seafloor. The maximum range of the

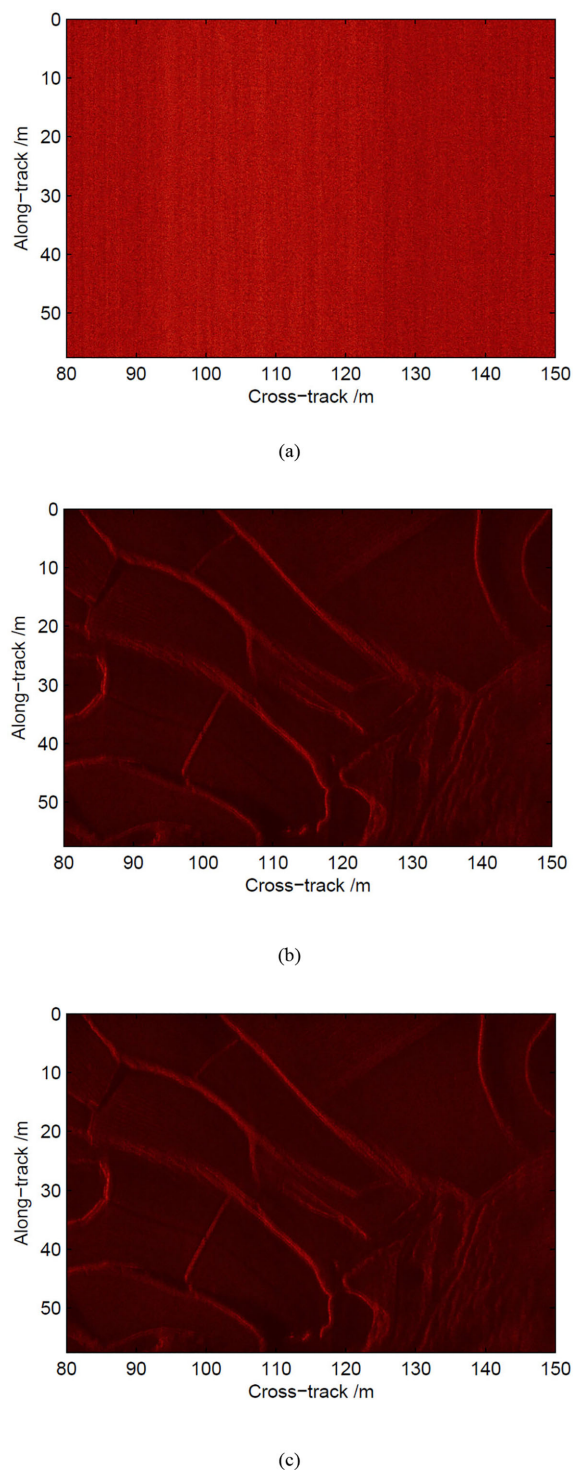


FIGURE 10
Reconstructed image using a real SAS system. (a) Traditional MI method, (b) Proposed method, and (c) the BP algorithm.

TABLE 2 Imagery metrics for the second simulation scenario.

Methods	Close target			Far target		
	PSLR (dB)	ISLR (dB)	RAT (m)	PSLR (dB)	ISLR (dB)	RAT (m)
MI method	—	—	—	—	—	—
Our method	-13.80	-10.51	0.04	-13.24	-9.64	0.04
BP algorithm	14.15	-10.71	0.04	-13.93	-9.97	0.04

TABLE 3 Processing times of real SAS datasets.

Time	MI method	Our method	BP algorithm
Processing time (s)	936	168	29813

TABLE 4 Contrast and entropy metrics of real SAS images.

Evaluation metrics	MI method	Our method	BP algorithm
Contrast	1.33	3.56	3.79
Entropy	6.72	4.52	4.43

active SAS is approximately several hundred meters. Too some degree, this maximum range is very close. Due to these reasons, the acoustic velocity is hardly affected by marine parameters such as temperature, and the proposed method exhibits robustness. However, the current study does not compensate for significant variance of acoustic velocity. In future, such variations will be compensated for in our work. In addition, further efforts will be directed toward optimizing the imaging algorithm described in this paper to further improve processing efficiency.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

LZ: Methodology, Investigation, Data curation, Supervision, Validation, Software, Writing – review & editing, Conceptualization, Resources, Visualization, Writing – original draft, Project administration, Formal analysis, Funding acquisition. XC: Funding acquisition, Writing – original draft, Formal analysis, Resources, Software, Visualization, Project administration, Supervision, Conceptualization, Methodology, Validation, Writing – review & editing, Data curation, Investigation. PW: Writing – original

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